



# Acetyltriethyl Citrate

## 1 Nonproprietary Names

USP-NF: Acetyltriethyl Citrate

## 2 Synonyms

ATEC; *Citroflex* A-2; triethyl acetylcitrate; triethyl O-acetylcitrate; triethyl citrate acetate.

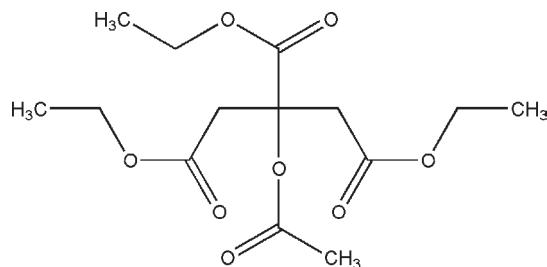
## 3 Chemical Name and CAS Registry Number

1,2,3-Propanetricarboxylic acid, 2-acetyloxy, triethyl ester [77-89-4]

## 4 Empirical Formula and Molecular Weight

C<sub>14</sub>H<sub>22</sub>O<sub>8</sub> 318.3

## 5 Structural Formula



## 6 Functional Category

Plasticizer.

## 7 Applications in Pharmaceutical Formulation or Technology

Acetyltriethyl citrate is used to plasticize polymers in formulated pharmaceutical coatings.<sup>(1)</sup> The coating applications include capsules, tablets, beads and granules for taste masking, immediate release, sustained-release and enteric formulations.<sup>(2-5)</sup> It is also used in diffusion-controlled release drug delivery systems.<sup>(6)</sup>

## 8 Description

Acetyltriethyl citrate occurs as a clear, odorless, practically colorless oily liquid.

## 9 Pharmacopeial Specifications

See Table I.

**Table I:** Pharmacopeial specifications for acetyltriethyl citrate.

| Test                    | USP32-NF27  |
|-------------------------|-------------|
| Identification          | +           |
| Specific gravity        | 1.135–1.139 |
| Refractive index        | 1.432–1.441 |
| Acidity                 | +           |
| Water                   | ≤0.3%       |
| Heavy metals            | ≤0.001%     |
| Assay (anhydrous basis) | ≥99.0%      |

## 10 Typical Properties

*Acid value* 0.02

*Boiling point* 294°C (decomposes)

*Flash point* 188°C

*Pour point* –43°C

*Solubility* Soluble 1 in 140 of water; miscible with acetone, ethanol, and propan-2-ol.

*Viscosity (dynamic)* 54 mPa s (54 cP) at 25°C.

## 11 Stability and Storage Conditions

Acetyltriethyl citrate should be stored in dry, closed containers at temperatures not exceeding 38°C. When stored in accordance with these conditions, acetyltriethyl citrate is a stable product.

## 12 Incompatibilities

Acetyltriethyl citrate is incompatible with strong alkalis and oxidizing materials.

## 13 Method of Manufacture

Acetyltriethyl citrate is prepared by the esterification of citric acid with ethanol followed by acylation with acetic anhydride.

## 14 Safety

Acetyltriethyl citrate is used in oral pharmaceutical formulations and is generally regarded as a nontoxic and nonirritating material. However, ingestion of large quantities may be harmful.

LD<sub>50</sub> (cat, oral): 8.5 g/kg<sup>(7)</sup>

LD<sub>50</sub> (mouse, IP): 1.15 g/kg

LD<sub>50</sub> (rat, oral): 7 g/kg

## 15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Acetyltriethyl citrate may be irritating to the eyes or the respiratory system as a mist or at elevated temperatures. Gloves and eye protection are recommended for normal handling and a respirator is recommended if used at elevated temperatures.

## 16 Regulatory Status

Approved in the USA for direct food contact in food films.

## 17 Related Substances

Acetyltributyl citrate; tributyl citrate; triethyl citrate.

## 18 Comments

The EINECS number for acetyltriethyl citrate is 201-066-5. The PubChem Compound ID (CID) for acetyltriethyl citrate is 6504.

## 19 Specific References

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- Gutierrez-Rocca JC, McGinity JW. Influence of water soluble and insoluble plasticizer on the physical and mechanical properties of acrylic resin copolymers. *Int J Pharm* 1994; 103: 293–301.
- Lehmann K. Chemistry and application properties of polymethacrylate coating systems. McGinity JW, ed. *Aqueous Polymeric Coatings for*

*Pharmaceutical Dosage Forms*. New York: Marcel Dekker, 1989; 153–245.

- 4 Steurnagel CR. Latex emulsions for controlled drug delivery. McGinity JW, ed. *Aqueous Polymeric Coatings for Pharmaceutical Dosage Forms*. New York: Marcel Dekker, 1–61.
- 5 Gutierrez-Rocca JC, McGinity JW. Influence of aging on the physical-mechanical properties of acrylic resin films cast from aqueous dispersions and organic solutions. *Drug Dev Ind Pharm* 1993; 19(3): 315–332.
- 6 Siepmann J *et al.* Diffusion-controlled drug delivery systems: calculation of the required composition to achieve desired release profiles. *J Control Release* 1999; 60(2–3): 379–389.
- 7 Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 58–59.

## 20 General References

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## 21 Authors

ME Quinn, PJ Sheskey.

## 22 Date of Revision

18 February 2009.

# Adipic Acid

## 1 Nonproprietary Names

PhEur: Adipic Acid

USP-NF: Adipic Acid

## 2 Synonyms

Acidum adipicum; acifloctin; acinetten; adilactetten; asapic; 1,4-butanedicarboxylic acid; E355; 1,6-hexanedioic acid; *Inipol DS*.

## 3 Chemical Name and CAS Registry Number

Hexanedioic acid [124-04-9]

## 4 Empirical Formula and Molecular Weight

C<sub>6</sub>H<sub>10</sub>O<sub>4</sub> 146.14

## 5 Structural Formula



## 6 Functional Category

Acidifying agent; buffering agent; flavoring agent.

## 7 Applications in Pharmaceutical Formulation or Technology

Adipic acid is used as an acidifying and buffering agent in intramuscular, intravenous and vaginal formulations. It is also used in food products as a leavening, pH-controlling, or flavoring agent.

Adipic acid has been incorporated into controlled-release formulation matrix tablets to obtain pH-independent release for both weakly basic<sup>(1,2)</sup> and weakly acidic drugs.<sup>(3,4)</sup> It has also been incorporated into the polymeric coating of hydrophilic monolithic systems to modulate the intragel pH, resulting in zero-order release

of a hydrophilic drug.<sup>(5)</sup> The disintegration at intestinal pH of the enteric polymer shellac has been reported to improve when adipic acid was used as a pore-forming agent without affecting release in the acidic media.<sup>(6)</sup> Other controlled-release formulations have included adipic acid with the intention of obtaining a late-burst release profile.<sup>(7)</sup>

## 8 Description

Adipic acid occurs as a white or almost white, odorless nonhygroscopic crystalline powder. The crystal structure of adipic acid is monoclinic holohedral.

## 9 Pharmacopeial Specifications

See Table I.

**Table I:** Pharmacopeial specifications for adipic acid.

| Test                   | PhEur 6.0   | USP32-NF27  |
|------------------------|-------------|-------------|
| Identification         | +           | +           |
| Characters             | +           | —           |
| Melting range          | 151–154°C   | 151–154°C   |
| Appearance of solution | +           | —           |
| Loss on drying         | ≤ 0.2%      | ≤ 0.2%      |
| Residue on ignition    | —           | ≤ 0.1%      |
| Sulfated ash           | ≤ 0.1%      | —           |
| Chlorides              | ≤ 200 ppm   | ≤ 0.02%     |
| Nitrates               | ≤ 30 ppm    | ≤ 0.003%    |
| Sulfates               | ≤ 500 ppm   | ≤ 0.05%     |
| Iron                   | ≤ 10 ppm    | ≤ 0.001%    |
| Heavy metals           | ≤ 10 ppm    | ≤ 0.001%    |
| Related substances     | +           | —           |
| Assay (anhydrous)      | 99.0–101.0% | 99.0–101.0% |

## 10 Typical Properties

**Acidity/alkalinity** pH = 2.7 (saturated solution at 25°C); pH = 3.2 (0.1% w/v aqueous solution at 25°C)

**Boiling point** 337.5°C

**Dissociation constant**

pK<sub>a1</sub>: 4.418 at 25°C;

pK<sub>a2</sub>: 5.412 at 25°C.

**Density** 1.360 g/cm<sup>3</sup>

**Flash point** 196°C (closed cup)